



Seeing sounds

You can show that sound vibrations exist by hitting a tray next to a drum sprinkled with rice. The rice bounces up and down with the vibrations that the sound makes.

Hitting a tin pan makes it vibrate. You hear a loud and crashing noise.

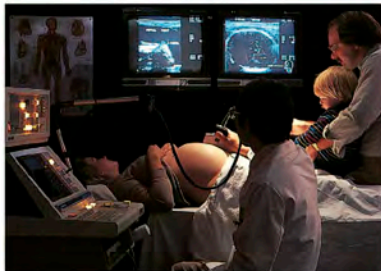


What is frequency?

The number of complete sound waves that pass by in a second gives us the "frequency" of a sound. Frequency is measured in hertz. High notes have a high frequency. They make lots of vibrations, and have a high number of hertz. Low notes have a low frequency.

When a wind instrument is played, the air inside the tube vibrates to make a sound.

Blowing into the pipe makes the air inside it vibrate all the way along to the funnel at the end.



Bouncing sound

This pregnant woman can see her baby before it is born because of sound. High-frequency sound waves are reflected as they reach the baby, making an image on the computer screen.

Hearing sounds

We cannot hear sounds of very high or low frequencies, but some animals can. It is impossible for us to hear these sounds because they are outside our hearing range.



Bat

up to 120,000 hertz



Mouse

up to 91,000 hertz



Cat

up to 64,000 hertz



Dog

up to 45,000 hertz



Young person

up to 20,000 hertz